Deep Dive 01 | Routes and Request Flow

Trace startup, browser navigation, proxying, Express middleware and API contracts.

React/Vite Edition | Repository-grounded study material | Snapshot: 2026-10-24

1. Web application entry

frontend/src/main.jsx mounts the React application and router providers; frontend/src/App.jsx declares routes. React Router maps the URL path to page components. Vite serves the development application and provides a proxy for API and upload paths.

2. Express startup

- dotenv loads backend configuration before database, Passport and routes initialize.
- connectDB starts a Mongoose connection using MONGO_URI; the server logs a warning rather than exiting if connection fails.
- CORS, JSON parsing, express-session, Passport initialization and session support are installed before route mounts.
- Auth, nearby and alert routers are mounted under /api; user and upload routes are mounted separately.
- The service exposes uploads, serves frontend/dist where available, and sends index.html for frontend routes.

3. Development and production request routing

Environment	Frontend request path
Local development	Browser -> Vite localhost:5173 -> Vite proxy /api or /uploads -> Express localhost:5000.
Vercel frontend	Browser -> Vercel static assets -> rewrite /api and /uploads to Render Express.
Single Node deployment	Express serves frontend/dist and handles /api in one process.
Chatbot	Browser -> Flask service URL from VITE_FLASK_URL -> /chat.

4. Express API map

Prefix/path	Owner	Contract summary
/api/auth/register	authRoutes	Multipart registration; optional picture; user plus JWT.
/api/auth/login	authRoutes	JSON email/password; user plus JWT.
/api/auth/google	Passport	OAuth authorize and callback.
/api/users/profile	userRoutes	Protected profile lookup.
/api/alerts	alertRoutes	NDMA alert summaries.
/api/alerts/cap/:id	alertRoutes	CAP XML detail normalized to JSON.
/api/nearby	nearbyRoutes	Coordinate-based place lookup with normalized output.
/api/upload	uploadRoutes	Profile image upload.

5. Middleware ordering matters

CORS and body parsing must run before routes that depend on them. Passport must be initialized before OAuth routes. The session middleware is configured because Passport sessions are enabled. Static file serving should not accidentally intercept API requests. The SPA fallback should run after API routes, or API errors could return HTML instead of JSON.

6. Study exercises

- Trace one GET /api/users/profile request from the React helper to the auth middleware and model query.
- Explain the effect of the Vite proxy on browser CORS during local development.
- Explain why a Vercel SPA needs a fallback rewrite for a deep route refresh.
- Identify what should happen if MongoDB connection fails before a profile request.